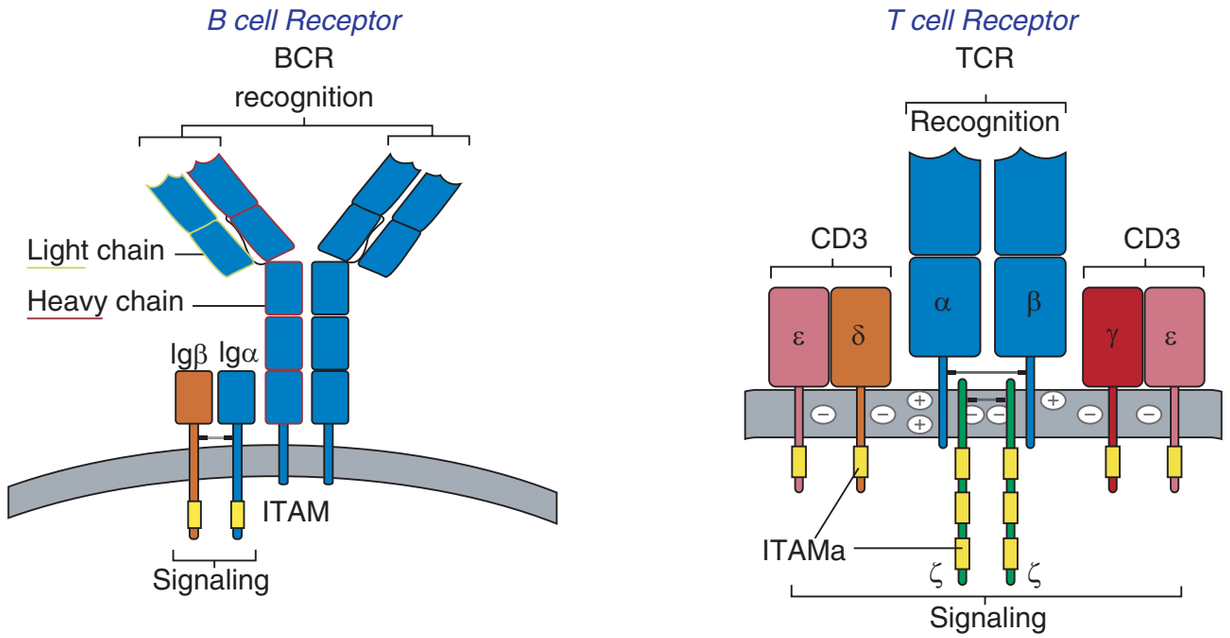


Figure 13.13: (a) B cell receptor. (b) T cell receptor

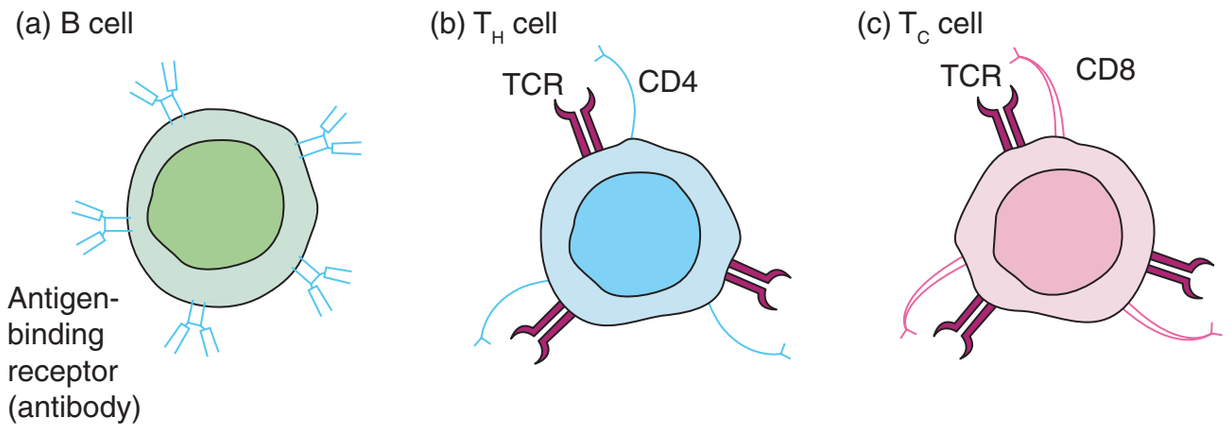


Figure 13.14: Distinctive membrane molecules on lymphocytes

macrophages and various other cells that participate in the immune response.

Under the influence of TH derived cytokines, a TC cell that recognizes an antigen-MHC class I molecule complex proliferates and differentiates into a cytotoxic T lymphocyte (CTL). Cells that display foreign antigen complexed with a class I MHC molecule are called altered self cells. CTL destroy virus infected cells and tumor cells.

iii) Natural killer (NK) Cells (Null cells)

NK cells are a small population of large, non phagocytic granular lymphocytes that play an important role in innate immunity. The major NK cell function is to destroy cancer cells and cells infected with microorganisms. They recognize their targets in one of two ways. They can bind to antibodies that coat infected or cancer cells. Thus the antibody bridges the two cell types. This process is called antibody dependent cell mediated cytotoxicity (ADCC) (Figure 13.15) The second way that NK cells recognize infected cells and cancer cells relies on the presence of specialized proteins on the surface of

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